Finite Element Method

2D Elasticity Problems

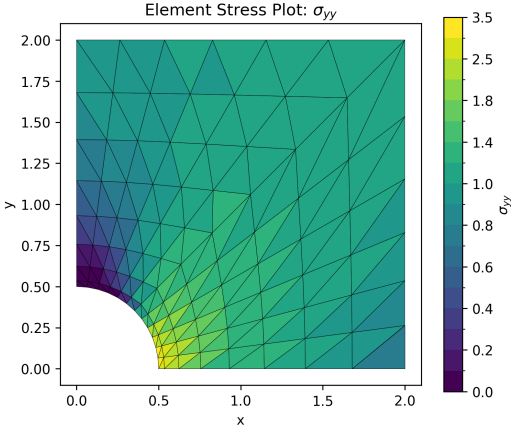
MMAE 450

Finite Element Method for Solid Mechanics

- Plane stress / plane strain formulation
- Weak form
- Linear triangular elements
- Assembly of the global stiffness matrix

Goal: derive the matrix system

$$\mathbf{K}\mathbf{d} = \mathbf{F}$$



Equilibrium Equations

Static equilibrium in a solid:

$$\nabla \cdot \boldsymbol{\sigma} + \mathbf{b} = 0 \qquad \text{in } \Omega$$

Boundary conditions:

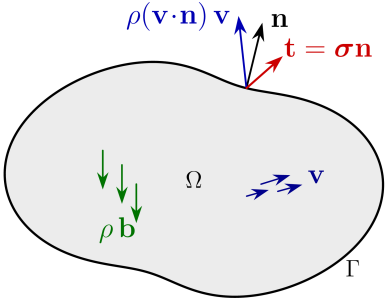
$$u_i = u_i^* \qquad \text{on } \Gamma_u$$

$$t_i = t_i^* \qquad \text{on } \Gamma_t$$

where

$$t_i = \sigma_{ij} n_j$$

- Γ_u: displacement boundary
- Γ_t: traction boundary



Momentum balance over an arbitrary region

Weak Form

Multiply the equilibrium equation by a test function w_i:

$$\int_\Omega w_i(\sigma_{ij,j} + b_i)\, d\Omega = 0$$

Test function requirements:

- $w_i = 0$ on Γ_u
- otherwise w_i has the same continuity as the displacement field

Such test functions are called **kinematically admissible**.

Weak Form

Applying the divergence theorem gives

$$\int_\Omega \sigma_{ij} w_{i,j} \, d\Omega = \int_\Gamma t_i w_i \, d\Gamma + \int_\Omega b_i w_i \, d\Omega$$

Using the boundary conditions:

$$\int_\Omega \sigma_{ij} w_{i,j} \, d\Omega = \int_{\Gamma_t} t_i^* w_i \, d\Gamma + \int_\Omega b_i w_i \, d\Omega$$

because $w_i = 0$ on Γ_u.

Product Rule Identity

To transform the integral statement we use the identity

$$(\sigma_{ij} w_i)_{,j} = \sigma_{ij,j} w_i + \sigma_{ij} w_{i,j}.$$

Rearranging,

$$\sigma_{ij,j} w_i = (\sigma_{ij} w_i)_{,j} - \sigma_{ij} w_{i,j}.$$

This identity shifts the derivative from the stress field to the test function.

This step reduces the continuity requirement on σ_{ij}.

Applying the Identity

Start with the statement

$$\int_\Omega w_i(\sigma_{ij,j} + b_i)\, d\Omega = 0.$$

Substitute

$$\sigma_{ij,j} w_i = (\sigma_{ij} w_i)_{,j} - \sigma_{ij} w_{i,j}$$

to obtain

$$\int_\Omega (\sigma_{ij} w_i)_{,j} d\Omega - \int_\Omega \sigma_{ij} w_{i,j} d\Omega + \int_\Omega b_i w_i d\Omega = 0.$$

Weak Form of the Equilibrium Equations

Applying the divergence theorem gives

$$\int_\Omega \sigma_{ij} w_{i,j} \, d\Omega = \int_\Gamma t_i w_i \, d\Gamma + \int_\Omega b_i w_i \, d\Omega$$

Applying the boundary conditions on $\Gamma = \Gamma_u + \Gamma_t$:

$$w_i = 0 \quad \text{on } \Gamma_u$$

gives

$$\boxed{\int_\Omega \sigma_{ij} w_{i,j} \, d\Omega = \int_{\Gamma_t} t_i^* w_i \, d\Gamma + \int_\Omega b_i w_i \, d\Omega}$$

Finite Element Approximation

The weak form involves two unknown fields:

$$u_i(x, y) \qquad \text{and} \qquad w_i(x, y)$$

In the finite element method we approximate these fields using shape functions.

Displacement approximation

$$u_i^e(x, y) = N_a(x, y)\, u_i^a$$

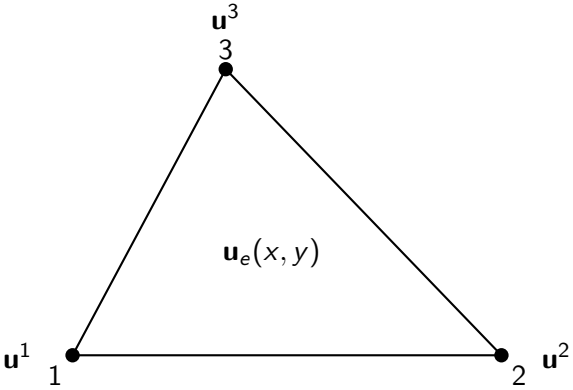
Test function approximation

$$w_i^e(x, y) = N_a(x, y)\, w_i^a$$

- $N_a(x, y)$: element shape functions
- u_i^a : nodal displacement components
- w_i^a : nodal values of the test function

Using the same shape functions for both fields is called the **Galerkin method**.

Three-Node Triangular Element



For a linear triangular element, the displacement field inside the element is interpolated from the nodal displacement vectors

$$\mathbf{u}^1, \qquad \mathbf{u}^2, \qquad \mathbf{u}^3.$$

Matrix Form of the Displacement Field

The nodal displacement vectors are stored in the element displacement vector

$$\underset{6\times 1}{\mathbf{d}_e} = \begin{bmatrix} u_x^1 \\ u_y^1 \\ u_x^2 \\ u_y^2 \\ u_x^3 \\ u_y^3 \end{bmatrix}$$

The displacement field inside the element is then written as

$$\mathbf{u}_e(x, y) = \mathbf{N}_e(x, y)\mathbf{d}_e$$

where

$$\mathbf{N}_e = \begin{bmatrix} N_1 & 0 & N_2 & 0 & N_3 & 0 \\ 0 & N_1 & 0 & N_2 & 0 & N_3 \end{bmatrix}.$$

Internal Work Term for Plane Stress

For plane stress, the only nonzero stress components are

$$\sigma_{11}, \qquad \sigma_{22}, \qquad \sigma_{12} = \sigma_{21}.$$

Thus

$$\sigma_{ij} w_{i,j} = \sigma_{11} w_{1,1} + \sigma_{22} w_{2,2} + \sigma_{12}(w_{1,2} + w_{2,1}).$$

Define the virtual strain vector

$$\delta \boldsymbol{\varepsilon}_e = \begin{bmatrix} w_{1,1} \\ w_{2,2} \\ w_{1,2} + w_{2,1} \end{bmatrix}, \qquad \boldsymbol{\sigma}_e = \begin{bmatrix} \sigma_{11} \\ \sigma_{22} \\ \sigma_{12} \end{bmatrix}.$$

Then

$$\sigma_{ij} w_{i,j} = \delta \boldsymbol{\varepsilon}_e^T \boldsymbol{\sigma}_e.$$

Hooke's Law for Plane Stress

Hooke's law for plane stress may be written as

$$\begin{bmatrix} \sigma_{11} \\ \sigma_{22} \\ \sigma_{12} \end{bmatrix} = \frac{E}{1-\nu^2} \begin{bmatrix} 1 & \nu & 0 \\ \nu & 1 & 0 \\ 0 & 0 & \dfrac{1-\nu}{2} \end{bmatrix} \begin{bmatrix} \epsilon_{11} \\ \epsilon_{22} \\ \gamma_{12} \end{bmatrix}$$

or

$$\boldsymbol{\sigma}_e = \mathbf{C}_e \boldsymbol{\varepsilon}_e$$

where

$$\boldsymbol{\sigma}_e = \begin{bmatrix} \sigma_{11} \\ \sigma_{22} \\ \sigma_{12} \end{bmatrix}, \qquad \boldsymbol{\varepsilon}_e = \begin{bmatrix} \epsilon_{11} \\ \epsilon_{22} \\ \gamma_{12} \end{bmatrix}, \qquad \mathbf{C}_e = \frac{E}{1-\nu^2} \begin{bmatrix} 1 & \nu & 0 \\ \nu & 1 & 0 \\ 0 & 0 & \dfrac{1-\nu}{2} \end{bmatrix}$$

Strain–Displacement Relations

The strain components are related to the displacement field

$$\mathbf{u}(x, y) = \begin{bmatrix} u_1(x, y) \\ u_2(x, y) \end{bmatrix}.$$

For small deformations,

$$\epsilon_{11} = \frac{\partial u_1}{\partial x_1}, \qquad \epsilon_{22} = \frac{\partial u_2}{\partial x_2}, \qquad \gamma_{12} = \frac{\partial u_1}{\partial x_2} + \frac{\partial u_2}{\partial x_1}.$$

Collecting these into vector form gives

$$\boldsymbol{\varepsilon} = \begin{bmatrix} \epsilon_{11} \\ \epsilon_{22} \\ \gamma_{12} \end{bmatrix} = \begin{bmatrix} u_{1,1} \\ u_{2,2} \\ u_{1,2} + u_{2,1} \end{bmatrix}.$$

Strain and Virtual Strain

Recall the strain vector

$$\boldsymbol{\varepsilon} = \begin{bmatrix} \epsilon_{11} \\ \epsilon_{22} \\ \gamma_{12} \end{bmatrix} = \begin{bmatrix} u_{1,1} \\ u_{2,2} \\ u_{1,2} + u_{2,1} \end{bmatrix}.$$

Similarly, the virtual strain associated with the test function is

$$\delta\boldsymbol{\varepsilon} = \begin{bmatrix} w_{1,1} \\ w_{2,2} \\ w_{1,2} + w_{2,1} \end{bmatrix}.$$

Notice the identical structure of these two expressions.

Using the finite element approximations,

$$\boxed{\boldsymbol{\varepsilon}_e = \mathbf{B}_e \mathbf{d}_e}, \qquad \boxed{\delta\boldsymbol{\varepsilon}_e = \mathbf{B}_e \mathbf{w}_e}.$$

Internal Virtual Work for a Single Element

For a single element Ω_e, the internal virtual work term is

$$\int_{\Omega_e} \sigma_{ij}\, w_{i,j}\, d\Omega = \int_{\Omega_e} \delta\boldsymbol{\varepsilon}_e^T \boldsymbol{\sigma}_e\, d\Omega$$

Using

$$\boldsymbol{\sigma}_e = \mathbf{C}_e \boldsymbol{\varepsilon}_e, \qquad \boldsymbol{\varepsilon}_e = \mathbf{B}_e \mathbf{d}_e, \qquad \delta\boldsymbol{\varepsilon}_e = \mathbf{B}_e \mathbf{w}_e,$$

we obtain

$$= \int_{\Omega_e} (\mathbf{B}_e \mathbf{w}_e)^T \mathbf{C}_e (\mathbf{B}_e \mathbf{d}_e)\, d\Omega$$

$$= \mathbf{w}_e^T \left(\int_{\Omega_e} \mathbf{B}_e^T \mathbf{C}_e \mathbf{B}_e\, d\Omega \right) \mathbf{d}_e.$$

Strain–Displacement Matrix for a Linear Triangle

The strain vector is

$$\boldsymbol{\varepsilon}_e = \begin{bmatrix} \epsilon_{11} \\ \epsilon_{22} \\ \gamma_{12} \end{bmatrix} = \begin{bmatrix} u_{1,1} \\ u_{2,2} \\ u_{1,2} + u_{2,1} \end{bmatrix} = \mathbf{B}_e \mathbf{d}_e,$$

where

$$\mathbf{B}_e = \begin{bmatrix} N_{1,1} & 0 & N_{2,1} & 0 & N_{3,1} & 0 \\ 0 & N_{1,2} & 0 & N_{2,2} & 0 & N_{3,2} \\ N_{1,2} & N_{1,1} & N_{2,2} & N_{2,1} & N_{3,2} & N_{3,1} \end{bmatrix}.$$

For the linear triangle, the derivatives of the shape functions are constant, so $\mathbf{B}_e$ is constant within the element.

Body Force Contribution

The body force term in the weak form is

$$\int_{\Omega_e} w_i b_i \, d\Omega.$$

Using the finite element approximation for the test function

$$\mathbf{w}(x, y) = \mathbf{N}_e(x, y)\mathbf{w}_e,$$

we obtain

$$\int_{\Omega_e} w_i b_i \, d\Omega = \int_{\Omega_e} (\mathbf{N}_e \mathbf{w}_e)^T \mathbf{b} \, d\Omega.$$

Rearranging gives

$$= \mathbf{w}_e^T \left(\int_{\Omega_e} \mathbf{N}_e^T \mathbf{b} \, d\Omega \right).$$

Traction Boundary Contribution

The traction term in the weak form is

$$\int_{\Gamma_t^e} w_i t_i^* \, d\Gamma.$$

Using the finite element approximation

$$\mathbf{w}(x, y) = \mathbf{N}_e(x, y)\mathbf{w}_e,$$

we obtain

$$\int_{\Gamma_t^e} w_i t_i^* \, d\Gamma = \int_{\Gamma_t^e} (\mathbf{N}_e \mathbf{w}_e)^T \mathbf{t}^* \, d\Gamma.$$

Rearranging gives

$$= \mathbf{w}_e^T \left(\int_{\Gamma_t^e} \mathbf{N}_e^T \mathbf{t}^* \, d\Gamma \right).$$

Total Element Contribution to the Weak Form

Substituting all of the finite element approximations into the weak form gives

$$\mathbf{w}_e^T \left(\int_{\Omega_e} \mathbf{B}_e^T \mathbf{C}_e \mathbf{B}_e \, d\Omega \right) \mathbf{d}_e = \mathbf{w}_e^T \left(\int_{\Omega_e} \mathbf{N}_e^T \mathbf{b} \, d\Omega + \int_{\Gamma_t^e} \mathbf{N}_e^T \mathbf{t}^* \, d\Gamma \right).$$

This equation represents the contribution of a single element to the global weak form.

Element Stiffness Matrix and Load Vector

Define the element stiffness matrix

$$\boxed{\mathbf{K}_e = \int_{\Omega_e} \mathbf{B}_e^T \mathbf{C}_e \mathbf{B}_e \, d\Omega}$$

and the element load vector

$$\boxed{\mathbf{f}_e = \int_{\Omega_e} \mathbf{N}_e^T \mathbf{b} \, d\Omega + \int_{\Gamma_t^e} \mathbf{N}_e^T \mathbf{t}^* \, d\Gamma}.$$

The element equation becomes

$$\mathbf{w}_e^T \mathbf{K}_e \mathbf{d}_e = \mathbf{w}_e^T \mathbf{f}_e.$$

Assembly of the Global System

The weak form must hold for all elements in the mesh.
Summing the element contributions gives

$$\sum_e \mathbf{w}_e^T \mathbf{K}_e \mathbf{d}_e = \sum_e \mathbf{w}_e^T \mathbf{f}_e.$$

Using Boolean assembly matrices

$$\mathbf{d}_e = \mathbf{L}_e \mathbf{d}, \qquad \mathbf{w}_e = \mathbf{L}_e \mathbf{w},$$

we obtain

$$\mathbf{w}^T \left(\sum_e \mathbf{L}_e^T \mathbf{K}_e \mathbf{L}_e \right) \mathbf{d} = \mathbf{w}^T \left(\sum_e \mathbf{L}_e^T \mathbf{f}_e \right).$$

Global Finite Element Equations

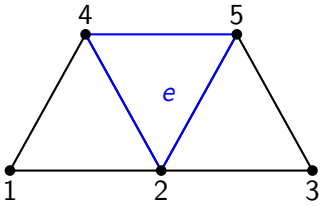
Since the equation must hold for all admissible $\mathbf{w}$,

$$\boxed{\mathbf{K}\mathbf{d} = \mathbf{F}}$$

where

$$\mathbf{K} = \sum_e \mathbf{L}_e^T \mathbf{K}_e \mathbf{L}_e, \qquad \mathbf{F} = \sum_e \mathbf{L}_e^T \mathbf{f}_e.$$

Boolean Matrix and Element Assembly



Element e connects nodes $(2, 4, 5)$.

The global displacement vector is

$$\mathbf{d} = \begin{bmatrix} u_1^1 & u_2^1 & u_1^2 & u_2^2 & u_1^3 & u_2^3 & u_1^4 & u_2^4 & u_1^5 & u_2^5 \end{bmatrix}^T .$$

For element $e = (2, 4, 5)$, the element displacement vector is

$$\mathbf{d}_e = \begin{bmatrix} u_1^2 & u_2^2 & u_1^4 & u_2^4 & u_1^5 & u_2^5 \end{bmatrix}^T .$$

Thus,

$$\boxed{\mathbf{d}_e = \mathbf{L}_e \, \mathbf{d}}$$

where $\mathbf{L}_e$ is the Boolean matrix that selects the degrees of freedom associated with element e.

Boolean Matrix Mapping

The element displacement vector is obtained from the global vector by

$$\mathbf{d}_e = \mathbf{L}_e \mathbf{d}.$$

For element $e = (2, 4, 5)$,

$$\begin{bmatrix} u_1^2 \\ u_2^2 \\ u_1^4 \\ u_2^4 \\ u_1^5 \\ u_2^5 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} u_1^1 \\ u_2^1 \\ u_1^2 \\ u_2^2 \\ u_1^3 \\ u_2^3 \\ u_1^4 \\ u_2^4 \\ u_1^5 \\ u_2^5 \end{bmatrix}.$$

Shape Functions for the Triangle (CST Element)

For a three-node linear triangular element, the displacement field is

$$\mathbf{u}_e(x, y) = \mathbf{N}_e(x, y)\mathbf{d}_e$$

where the shape-function matrix is

$$\mathbf{N}_e = \begin{bmatrix} N_1 & 0 & N_2 & 0 & N_3 & 0 \\ 0 & N_1 & 0 & N_2 & 0 & N_3 \end{bmatrix}.$$

The functions $N_1(x, y), N_2(x, y), N_3(x, y)$ are the **linear shape functions** associated with the three nodes.

Linear Shape Functions for a Triangle

For a three-node triangular element, each shape function is a linear polynomial in x and y:

$$N_a(x, y) = \frac{1}{2A_e}\left(\alpha_a + b_a x + c_a y\right), \qquad a = 1, 2, 3.$$

The coefficients depend only on the nodal coordinates

$$(x_1, y_1), \qquad (x_2, y_2), \qquad (x_3, y_3).$$

The element area is

$$A_e = \frac{1}{2}\begin{vmatrix} 1 & x_1 & y_1 \\ 1 & x_2 & y_2 \\ 1 & x_3 & y_3 \end{vmatrix}.$$

Coefficients α_a

The constants α_a are

$$\alpha_1 = x_2 y_3 - x_3 y_2$$

$$\alpha_2 = x_3 y_1 - x_1 y_3$$

$$\alpha_3 = x_1 y_2 - x_2 y_1$$

These depend only on the element nodal coordinates.

Coefficients b_a and c_a

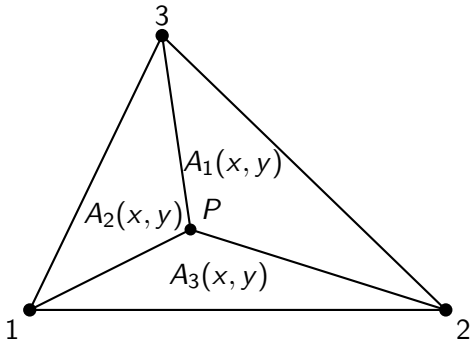
The remaining coefficients are

$$b_1 = y_2 - y_3, \qquad b_2 = y_3 - y_1, \qquad b_3 = y_1 - y_2$$

$$c_1 = x_3 - x_2, \qquad c_2 = x_1 - x_3, \qquad c_3 = x_2 - x_1$$

For a linear triangular element the derivatives of the shape functions are constant within the element.

Area Coordinate Interpretation



$$\boxed{N_1 + N_2 + N_3 = 1}$$

$$N_a(x_b, y_b) = \delta_{ab}$$

- Partition of unity
- Interpolation property

Let A_e be the total area of the element. The shape functions may be written as

$$N_1(x, y) = \frac{A_1(x, y)}{A_e}, \qquad N_2(x, y) = \frac{A_2(x, y)}{A_e}, \qquad N_3(x, y) = \frac{A_3(x, y)}{A_e}.$$

Derivatives of the Shape Functions

For a linear triangular element,

$$N_a(x, y) = \frac{1}{2A_e}\left(\alpha_a + b_a x + c_a y\right).$$

Taking derivatives gives

$$\frac{\partial N_a}{\partial x} = \frac{b_a}{2A_e}, \qquad \frac{\partial N_a}{\partial y} = \frac{c_a}{2A_e}.$$

Thus the derivatives of the shape functions are constant throughout the element.

This is why the element is called the

$$\boxed{\text{Constant Strain Triangle (CST)}}$$

Strain–Displacement Matrix $\mathbf{B}_e$

Recall that

$$\varepsilon_e = \begin{bmatrix} \epsilon_{11} \\ \epsilon_{22} \\ \gamma_{12} \end{bmatrix} = \begin{bmatrix} u_{1,1} \\ u_{2,2} \\ u_{1,2} + u_{2,1} \end{bmatrix}.$$

Using the finite element displacement approximation, we obtain

$$\varepsilon_e = \mathbf{B}_e \mathbf{d}_e$$

with

$$\mathbf{B}_e = \begin{bmatrix} N_{1,1} & 0 & N_{2,1} & 0 & N_{3,1} & 0 \\ 0 & N_{1,2} & 0 & N_{2,2} & 0 & N_{3,2} \\ N_{1,2} & N_{1,1} & N_{2,2} & N_{2,1} & N_{3,2} & N_{3,1} \end{bmatrix}.$$

$\mathbf{B}_e$ in Terms of Area and Nodal Coordinates

Since

$$N_{a,1} = \frac{b_a}{2A_e}, \qquad N_{a,2} = \frac{c_a}{2A_e},$$

we may write

$$\mathbf{B}_e = \frac{1}{2A_e} \begin{bmatrix} b_1 & 0 & b_2 & 0 & b_3 & 0 \\ 0 & c_1 & 0 & c_2 & 0 & c_3 \\ c_1 & b_1 & c_2 & b_2 & c_3 & b_3 \end{bmatrix},$$

Thus,

$$\mathbf{B}_e = \frac{1}{2A_e} \begin{bmatrix} y_2 - y_3 & 0 & y_3 - y_1 & 0 & y_1 - y_2 & 0 \\ 0 & x_3 - x_2 & 0 & x_1 - x_3 & 0 & x_2 - x_1 \\ x_3 - x_2 & y_2 - y_3 & x_1 - x_3 & y_3 - y_1 & x_2 - x_1 & y_1 - y_2 \end{bmatrix}.$$

Computing the CST Element Matrices

In the companion notebook we derive

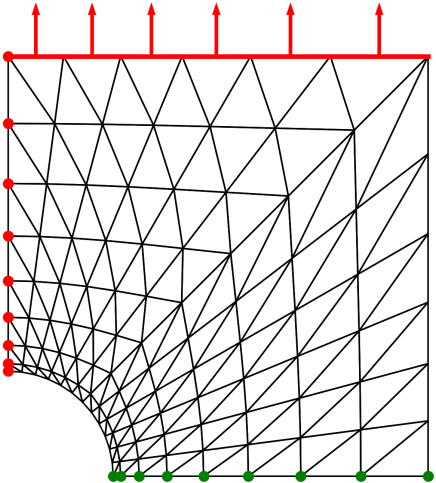
$$N_a(x, y) \quad \rightarrow \quad \mathbf{B}_e \quad \rightarrow \quad \mathbf{K}_e$$

for the three-node triangular element.

The notebook symbolically computes

$$\mathbf{B}_e = \frac{1}{2A_e} \begin{bmatrix} y_2 - y_3 & 0 & y_3 - y_1 & 0 & y_1 - y_2 & 0 \\ 0 & x_3 - x_2 & 0 & x_1 - x_3 & 0 & x_2 - x_1 \\ x_3 - x_2 & y_2 - y_3 & x_1 - x_3 & y_3 - y_1 & x_2 - x_1 & y_1 - y_2 \end{bmatrix}.$$

Example Problem Used in the Notebook



Geometry

- Plate with a circular hole
- Plane stress model

Boundary Conditions

- Left edge: $u_x = 0$
- Bottom edge: $u_y = 0$
- Right edge: traction free
- Top edge: $t_y = t_y^*, t_x = 0$

Material

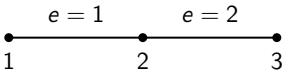
- Young's modulus E
- Poisson ratio ν

Finite Element Method for Solids: Algorithm Summary

Assumptions: small strains, linear elastic material, static equilibrium

1. Interpolate displacement: $\mathbf{u}_e = \mathbf{N}_e \mathbf{d}_e$
2. Strain–displacement relation: $\varepsilon_e = \mathbf{B}_e \mathbf{d}_e$
3. Constitutive law: $\boldsymbol{\sigma}_e = \mathbf{C}_e \varepsilon_e$
4. Element matrices: $\mathbf{K}_e = \int_{\Omega_e} \mathbf{B}_e^T \mathbf{C}_e \mathbf{B}_e \, d\Omega$, $\mathbf{f}_e = \int_{\Omega_e} \mathbf{N}_e^T \mathbf{b} \, d\Omega + \int_{\Gamma_t^e} \mathbf{N}_e^T \mathbf{t}^* \, d\Gamma$
5. Assemble: $\mathbf{K}\mathbf{d} = \mathbf{F}$
6. Solve for nodal displacements: $\mathbf{d}$
7. Post-process: $\varepsilon_e = \mathbf{B}_e \mathbf{d}_e$, $\boldsymbol{\sigma}_e = \mathbf{C}_e \varepsilon_e$

Assembly of the Global System



$$\mathbf{K}_1 = \begin{array}{cc} \begin{bmatrix} k_{11}^{(1)} & k_{12}^{(1)} \\ k_{21}^{(1)} & k_{22}^{(1)} \end{bmatrix} & \begin{array}{c} 1 \\ 2 \end{array} \\ \begin{array}{cc} 1 & 2 \end{array} & \end{array} \qquad \mathbf{K}_2 = \begin{array}{cc} \begin{bmatrix} k_{11}^{(2)} & k_{12}^{(2)} \\ k_{21}^{(2)} & k_{22}^{(2)} \end{bmatrix} & \begin{array}{c} 2 \\ 3 \end{array} \\ \begin{array}{cc} 2 & 3 \end{array} & \end{array}$$

$$\mathbf{K} = \begin{array}{cc} \begin{bmatrix} k_{11}^{(1)} & k_{12}^{(1)} & 0 \\ k_{21}^{(1)} & k_{22}^{(1)} + k_{11}^{(2)} & k_{12}^{(2)} \\ 0 & k_{21}^{(2)} & k_{22}^{(2)} \end{bmatrix} & \begin{array}{c} 1 \\ 2 \\ 3 \end{array} \\ \begin{array}{ccc} 1 & 2 & 3 \end{array} & \end{array}$$

For element e,

$$\mathbf{d}_e = \mathbf{L}_e \mathbf{d}$$

$$\mathbf{w}_e = \mathbf{L}_e \mathbf{w}$$

Element contributions are assembled into the global system as

$$\mathbf{K} = \sum_e \mathbf{L}_e^T \mathbf{K}_e \mathbf{L}_e$$

$$\mathbf{F} = \sum_e \mathbf{L}_e^T \mathbf{f}_e$$

Assembly simply places each element contribution into the correct global rows and columns.